

KARL HILL

AEROSPACE MISSION SOFTWARE
PLATFORM ENGINEERING | DEVSECOPS | TECHNICAL LEADERSHIP

SUMMARY

Staff Aerospace Software Engineer and technical lead with 20+ years building mission-critical software, cloud-native platforms, and high-assurance delivery systems across NASA, aerospace, and defense environments. Hands-on across cloud platforms, Kubernetes, CI/CD, and developer tooling — plus Python, DevSecOps, and cross-team software delivery.

SELECTED LEADERSHIP IMPACT

- Lead software delivery across mission systems, coordinating engineering execution, integration readiness, and releases across internal and external teams.
- Standardize CI/CD, developer tooling, and release practices across mission software repositories — pipelines, quality gates, and a repeatable path from commit to a constrained environment.
- Drive cloud-platform and Kubernetes delivery (containers, Helm/OCI packaging, multi-environment baselines) so teams ship the same way in isolated and integrated systems.
- Translate mission and operational requirements into engineering roadmaps, sequenced delivery plans, and executable software work.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs — National Security · Chantilly, VA

- Lead delivery of cloud-native mission simulation and telemetry for government aerospace and defense programs — CI/CD, multi-repo delivery, and release readiness in constrained environments.
- Guide team execution through onboarding, coaching, and delivery discipline, raising engineering quality, ownership, and accountability.
- Coordinate multi-environment release readiness across isolated and integrated systems, reducing late-stage integration risk and improving baseline traceability.
- Drive platform engineering and developer tooling — CI/CD, automated testing, quality gates, and security checks that shorten feedback loops in cloud-native environments.
- Define engineering governance — repository standards, PR/review policy, Definition of Done, and release process — across multi-repo, multi-environment systems.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Architected NASA's cloud-based Flood Mapping System on AWS, delivering near real-time, satellite-derived flood products to support disaster response.
- Rebuilt NASA Earth Observatory's high-traffic web platform, supporting ~1.5M monthly visitors while improving performance, UX, and SEO.
- Delivered an automated content registry workflow, boosting data collection efficiency by ~60% and accelerating researcher access to new datasets.
- Implemented Kubernetes delivery with GitLab CI/CD, Docker, and Helm — automated deployments, repeatable releases, and a consistent path through stakeholder approvals.
- Built a high-performance file and metadata platform on Ceph, improving virtual directory mapping and accelerating discovery for large scientific datasets.

DETAILS

Washington, DC

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LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

CORE COMPETENCIES

- Technical Leadership
- Platform Engineering & Developer Experience
- Cloud Platforms & Kubernetes
- CI/CD & DevSecOps
- Mission Software Delivery
- Systems Integration
- Engineering Governance

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, D.C.

- Architected and delivered a Laravel-based case management platform, reducing operational costs by \$30K/year.
- Led CRM enhancements that improved retention and contributed ~15% revenue growth through better lifecycle workflows and reporting.
- Spearheaded platform upgrades and security process improvements, doubling incident response efficiency and strengthening operational readiness.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, D.C.

- Delivered SugarCRM solutions for 20+ clients (including Virginia Department of Transportation, Washington Redskins, and Kastle Systems), improving sales operations and team productivity.
- Drove execution discipline that cut backlog ~90%, improving delivery predictability, product quality, and customer satisfaction.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal roles across travel, enterprise CRM, telecommunications, and managed security — delivering production systems for clients including Comcast, Mastercard, Verizon/MCI, and global travel customers.
- Shipped Finium, the multi-tenant managed-security platform that enabled a \$105M acquisition, and helped mature engineering practices around testing, code quality, and cross-functional delivery.

EDUCATION

Bachelor of Science in Computer Science coursework, University of Maryland

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe® Agilist, Scaled Agile (in progress)

Professional Scrum Master™ II, Scrum.org

Professional Scrum Product Owner™ I, Scrum.org

DEVELOPER TOOLING

bb-run — Local Bitbucket Pipelines runner — CI you can execute on a laptop.

pipeguard — Policy-as-code checks for CI/CD standards and deployment safety.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Kubernetes: AWS, Docker, Kubernetes, Helm/OCI

CI/CD & Developer Tooling: Bitbucket Pipelines, GitLab CI, GitHub Actions, Release Automation

Python Tooling: uv, Ruff, pytest, ty, pre-commit

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, Distributed Systems

Data: PostgreSQL, Elasticsearch/OpenSearch, Redis